Worlds Teach, Actions Execute: Decoupled World-to-Action Distillation for Robot Manipulation

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***Abstract*— World Action Models (WAMs) jointly predict future visual states and actions, inheriting rich spatiotemporal priors from video foundation models, yet requiring explicit video generation at every inference step introduces unnecessary computational cost for action prediction. Recent approaches accelerate WAMs by simplifying or bypassing video generation, but risk weakening the inherited physical priors or introducing additional processing modules. We advocate a different perspective: the world model should *teach* during training by providing structured future dynamics, while the action model independently *executes* at inference without generative overhead. Built on this principle, we introduce DAWN (Decoupled Action from World kNowledge), which connects a frozen world model and an action model through dynamics-appearance disentangled distillation, enabling the action model to internalize manipulation-critical future dynamics while filtering out irrelevant visual appearance. At inference, only the action model runs, achieving WAM-level performance without video prediction. Extensive experiments on LIBERO, LIBERO-Plus, RoboTwin 2.0, and real-world tasks show that DAWN consistently matches or surpasses state-of-the-art methods while eliminating inference-time video generation cost. Code will be released.**

Keywords: World Action Models, Knowledge Distillation, Robot Manipulation

I. Introduction

Robotic manipulation requires grounding visual observations and language instructions into executable actions, motivating Vision-Language-Action (VLA) models [1]–[3]. While benefiting from large-scale vision-language pretraining, their reliance on static image-text data limits exposure to temporal dynamics governing physical interaction. World Action Models (WAMs) [4]–[6] enrich policy learning with spatiotemporal foresight by jointly predicting future visual states and actions, inheriting rich priors from video foundation models [7], [8] and leveraging video prediction as visual planning [9], [10].

Existing WAMs follow two paradigms: cascaded architectures [8], [11], where a world branch synthesizes a visual plan before a separate action branch decodes commands; and joint architectures [4]–[6] that unify future state prediction and action generation. Both typically couple video prediction with action generation at inference, and iterative denoising for video synthesis introduces considerable computational overhead, challenging real-time closed-loop control.

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Recent work makes progress toward reconciling WAM effectiveness with inference efficiency. Fast-WAM [12] retains video co-training while removing explicit future generation at inference, repurposing the video backbone as a single-pass encoder. However, its design—where action tokens do not attend to future video tokens—limits the action model from fully leveraging future-predictive dynamics. S-VAM [13] compresses multi-step video denoising into single-step inference via self-distillation, yet still requires auxiliary modules at inference, adding overhead beyond a standalone action model. These observations suggest an ideal solution should enable the action model to inherit the world model's future-predictive knowledge while operating independently at inference.

As illustrated in Figure 1, we advocate a different perspective: the world model and action model should assume entirely *asymmetric* roles—the world model *teaches* during training by providing structured future dynamics as supervisory signal, while the action model independently *executes* at inference without generative overhead. This motivates three desiderata: **(i) Fast**—no video prediction or auxiliary processing at inference; **(ii) Informed**—the action model should encode future transition dynamics captured by the world model; and **(iii) Focused**—representations should prioritize action-relevant dynamics over visual appearance details.

A natural approach is to distill the world model's intermediate features into the action model via forward dynamics prediction. However, these features deeply entangle visual appearance—textures, lighting, backgrounds—with action-relevant dynamics—object displacements, gripper movements, contact changes. Aligning to such entangled features forces the action model to expend capacity on appearance information irrelevant for control, diluting the action-predictive signal.

Building on this, we introduce **DAWN** (**D**ecoupled **A**ction from **W**orld k**N**owledge), a framework connecting a dedicated world model and an action model through **dynamics-appearance disentangled distillation**. An auxiliary appearance generation pathway absorbs visual appearance from the world model's future features, compelling the action model to specialize in action-relevant dynamics critical for manipulation. During training, the frozen world model transfers *action-centric structural priors*; at inference, the world model and distillation module are discarded, and only the action model operates. Experiments demonstrate that DAWN achieves performance comparable to WAM-based approaches while eliminating inference-time video

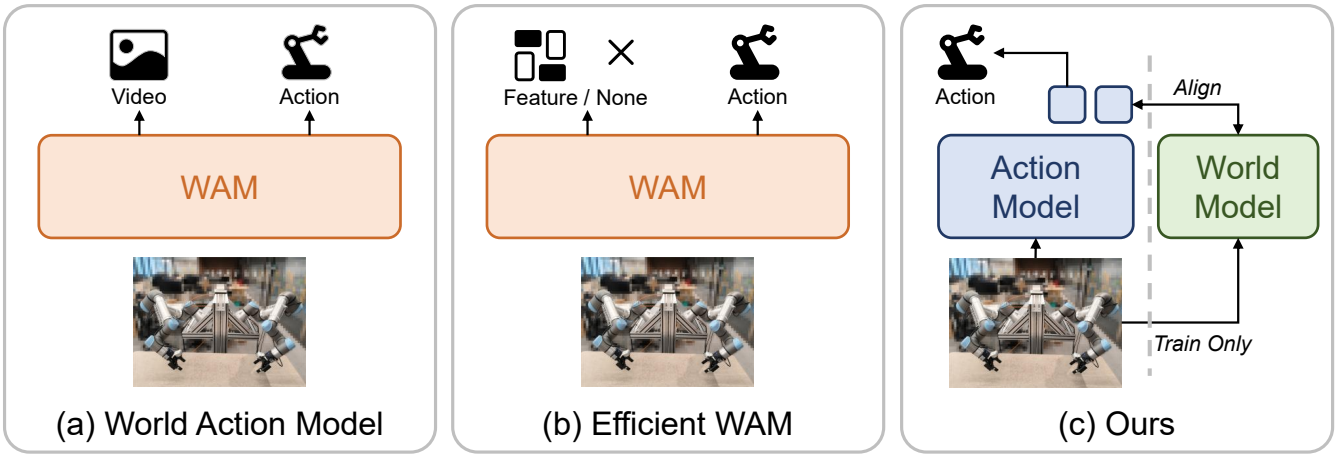


Fig. 1. **World Action Model inference paradigms.** (a) Standard WAMs synthesize explicit future video at inference, providing spatiotemporal context but incurring substantial cost from iterative denoising. (b) Efficient WAMs eliminate full video generation by repurposing the backbone as a feature encoder or compressing denoising, yet still retain auxiliary world-model components at inference. (c) *Ours*: the world model serves solely as a training-time teacher, distilling action-centric structural priors into the action model via dynamics-appearance disentangled distillation; at inference, only the standalone action model operates with no generative overhead.

generation cost.

Our contributions are summarized as follows:

- We identify the tension in World Action Models between inference efficiency and future-informed action prediction, and propose a training-inference asymmetric paradigm where the world model teaches during training and only the action model executes at inference.
- We introduce dynamics-appearance disentangled distillation, leveraging complementary learning with an auxiliary appearance pathway to ensure the action model internalizes action-relevant dynamics while filtering out irrelevant visual appearance.
- Experiments on LIBERO, LIBERO-Plus, RoboTwin 2.0, and real-world tasks demonstrate that DAWN achieves WAM-level performance with only the action model at inference, eliminating the computational overhead of video generation.

A comprehensive review of related work is provided in Appendix A.

II. Method

A. Problem Formulation and Overview

a) Task Paradigm: Given a visual observation o_t and a language instruction l, the goal is to predict an action sequence $a_{t:t+H}$ over an action horizon H. Robot manipulation tasks are challenging because the policy must ground high-level language goals in the current scene, identify task-relevant objects and spatial relations, and produce temporally coherent low-level controls under contact-rich interactions, occlusions, and changing object states.

b) Framework Overview: DAWN addresses this challenge through a training-inference asymmetry with three components, illustrated in Figure 2:

(1) A *frozen world model*—built on the LTX-Video [14] video diffusion transformer, pretrained on large-scale video data and adapted to multi-view robot observations, with parameters ϕ frozen after adaptation. Given o_t and l, it generates K future frames and provides intermediate layer features $\mathbf{f}_{t+1:t+K}$ as structural priors.

(2) A *dynamics-appearance disentangled distillation module* that selectively transfers action-centric priors from these features.

(3) An *action model* that predicts control commands independently of the video generation model. Since the world model is completely removed at inference time, the action model cannot rely on world-model features as inputs; it must encode the raw observation and instruction on its own while absorbing the structural priors provided during training. We therefore instantiate the model with the Qwen3-VL [15] vision-language backbone, whose strong capacity supports standalone vision-language understanding and the distillation of future-scene priors, and attach a lightweight MLP-based action head to predict $\hat{a}_{t:t+H}$.

During training, all three components interact; at inference, only the action model operates:

$$p_\theta(a_{t:t+H} \mid o_t, l) = p_\theta(a_{t:t+H} \mid \mathbf{z}_a(o_t, l)), \tag{1}$$

where $\mathbf{z}_a$ denotes the action-predictive embedding produced solely from the current observation, enriched during training with distilled action-centric structural priors.

B. Dynamics-Appearance Disentangled Distillation

The future features $\mathbf{f}_{t+1:t+K}$ entangle *action-relevant dynamics* (object displacements, gripper changes, contacts) with *visual appearance* (textures, lighting, backgrounds). Direct alignment to such entangled features wastes action representation capacity on control-irrelevant appearance. To address this, as illustrated in Figure 3, we introduce an auxiliary appearance generation pathway built upon Instruct-Pix2Pix [16], a conditional image editing diffusion model finetuned with LoRA [17]. Its pretrained image-to-image generation capacity naturally fits modeling frame-to-frame

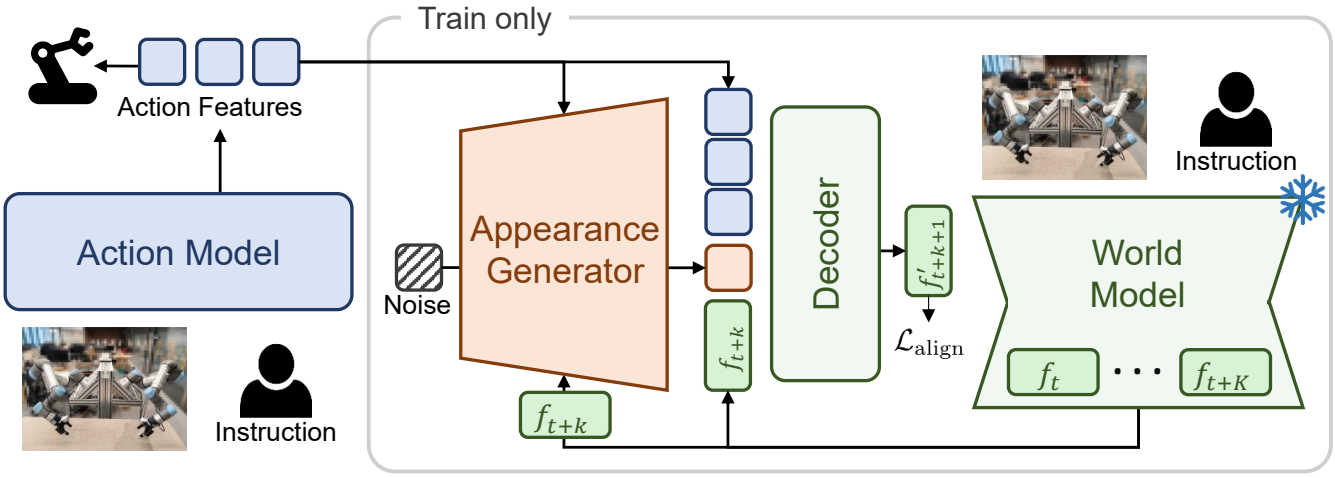


Fig. 2. **Overview of DAWN.** During training, a frozen world model generates future structural features from the current observation and language instruction. The dynamics-appearance disentangled distillation module separates action-relevant dynamics from visual appearance: an auxiliary appearance generator absorbs appearance information, while the action model's representations are driven to encode the complementary dynamics. At inference, the world model and distillation module are discarded; only the action model operates.

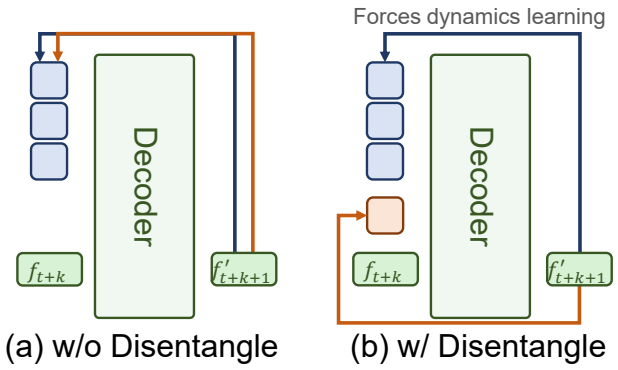


Fig. 3. **Dynamics-appearance disentangled distillation.** (a) Without disentanglement, directly aligning to entangled world-model features forces the action model to encode both dynamics and irrelevant appearance. (b) With disentanglement, the auxiliary appearance generator absorbs visual appearance, driving the action model to specialize in action-relevant dynamics.

appearance transitions, explicitly decoupling dynamics from appearance.

The world model's future features are organized into adjacent frame transition pairs following a forward dynamics formulation, modeling $\mathbf{f}_{t+k} \rightarrow \mathbf{f}_{t+k+1}$. The source frame features are projected into a compact latent space as the image condition, while the action model's intermediate representations Q are projected into the editing model's conditioning space via a learnable projector. The editing model generates visual appearance information for the future frame:

$$v_{t+k+1} = \mathscr{E}(\text{Proj}_{\text{img}}(\mathbf{f}_{t+k}),\ \text{Proj}_{\text{act}}(Q)), \tag{2}$$

where $\mathscr{E}$ is the pretrained editing model, Proj_{img} projects source features into latent space, and Proj_{act} maps action representations into the conditioning space. A decoder then reconstructs the next-frame structural features jointly conditioned on the source features, generated appearance, and action embeddings:

$$\hat{\mathbf{f}}_{t+k+1} = \text{Decode}(\mathbf{f}_{t+k},\ v_{t+k+1},\ Q). \tag{3}$$

Since the pretrained editing model inherently handles visual appearance, the action embeddings are driven to encode the complementary residual—action-relevant dynamics. This ensures the action representations specialize in task-relevant dynamics while the auxiliary pathway absorbs appearance.

C. Training and Inference

Stage 1: World Model Preparation. The world model is pretrained and finetuned as described above, after which the parameters ϕ are frozen. Full training data composition is provided in Appendix B-B.

Stage 2: Distillation and Action Model Training. The distillation module and the action model are trained end-to-end with the frozen world model providing structural supervision. The total training objective $\mathscr{L} = \mathscr{L}_{\text{act}} + \lambda \mathscr{L}_{\text{align}}$ combines a *structural prior alignment loss* $\mathscr{L}_{\text{align}}$ that aligns the reconstructed features with the world model's future structural features, and an *action prediction loss* $\mathscr{L}_{\text{act}}$ that supervises the final action output, where λ is a balancing coefficient.

Inference. The world model and distillation module are discarded. Only the action model runs:

$$a_{t:t+H} = \text{ActionModel}(o_t, l). \tag{4}$$

This eliminates the need for video generation at inference while retaining the action-centric structural priors distilled during training.

III. Experiments

A. Implementation Details

World model. The frozen world model is built on a 2B-parameter video DiT backbone (LTX-Video [14]) adapted to multi-view robot observations, providing intermediate

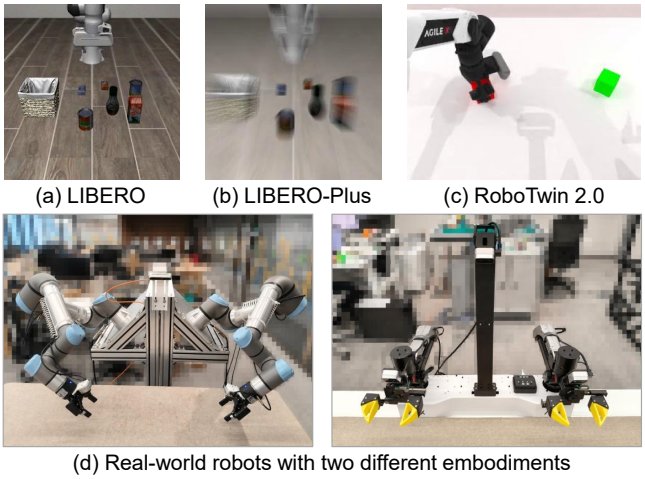


Fig. 4. **Evaluation benchmarks and robot platforms.** We evaluate DAWN on three simulation benchmarks—LIBERO, LIBERO-Plus, and RoboTwin 2.0—and on two real-world dual-arm embodiments used for physical manipulation experiments.

structural features as distillation targets. Details are in Appendix B.

Appearance generation pathway. The auxiliary appearance branch is built upon InstructPix2Pix [16], a conditional latent diffusion model finetuned with LoRA [17], that reconstructs appearance-level information from the world model's future predictions, driving the action branch to specialize in dynamics through complementary learning.

Action prediction module. A Qwen3-VL [15] backbone with a lightweight MLP action head independently encodes observations and instructions at deployment, as the world model and distillation module are discarded. Details are in Appendix C.

B. Simulation Benchmarks

We evaluate on three benchmarks: LIBERO [18] (single-arm tabletop), LIBERO-Plus [19] (robustness under domain perturbations), and RoboTwin 2.0 [20] (bimanual manipulation). We compare the full DAWN with two variants: DAWN w/ WM Infer., which retains the world model at inference to generate future video from the current observation for action prediction; and DAWN w/o WM Distill, which removes distillation entirely.

Benchmarks. *LIBERO* comprises four suites (Spatial, Object, Goal, Long) with 10 tasks each evaluated over 50 episodes. *LIBERO-Plus* extends LIBERO with perturbations along seven dimensions (Camera, Robot, Language, Light, Background, Noise, Layout); policies are trained on original LIBERO data without fine-tuning. *RoboTwin 2.0* evaluates coordinated dual-arm control on 10 jointly-trained tasks categorized by horizon length into short (≤ 400), medium (400–800), and long (≥ 800) steps. The benchmark suite and real-world platforms are summarized in Figure 4, and qualitative simulation rollouts are in Appendix E-A.

LIBERO. As shown in Table I, DAWN achieves the highest average success rate among prior baselines, surpassing OpenVLA-OFT [2], [23] and $\pi_{0.5}$ [28]. The improvement over DAWN w/o WM Distill validates that distilled structural priors contribute meaningful future-predictive information. Moreover, DAWN closely matches DAWN w/ WM Infer., confirming that our distillation preserves WAM-level performance without inference-time video generation.

LIBERO-Plus. Table II evaluates robustness under domain perturbations. Fast-WAM [12], despite incorporating world modeling via video co-training, enforces a strict information barrier preventing action tokens from attending to future tokens, yielding only 15.9% on Camera and 37.4% on Noise. DAWN directly distills action-centric structural priors, achieving the highest average among baselines and surpassing OpenVLA-OFT by 9.1 points. DAWN is also comparable to or even outperforms DAWN w/ WM Infer., confirming that distilled priors preserve generalization benefits without video generation.

RoboTwin 2.0. Table III shows DAWN achieves the highest average among baselines, outperforming DP3 [32] by 26.7 points with advantages most pronounced on long-horizon tasks. DAWN remains on par with DAWN w/ WM Infer., confirming that the distilled priors largely preserve the world model's spatiotemporal reasoning capacity. The gap to DAWN w/o WM Distill further shows that distillation becomes increasingly important with task complexity.

C. Real-World Experiments

Experiment setup. We deploy DAWN on two real robot platforms: a dual-arm UR5e system and a dual-arm ARX ACONE system, each equipped with three RGB cameras (one head-mounted and two wrist-mounted). The hardware embodiments and camera layouts are detailed in Figure 6 in Appendix D. We design six manipulation tasks spanning rigid-object insertion, deformable-object folding, and multi-step sequential manipulation: *Insert the Drill*, *Hang the Towel on the Hanger*, *Place the Chain into the Box and Close the Box*, *Fold the Cloth*, *Fold the Towel*, and *Stack the Blocks*; their initial and goal configurations are shown in Figure 7 in Appendix D. For each task, we collect 300 teleoperated demonstrations for training. We compare against two baselines: GR00T N1.5 [27], a powerful VLA foundation model pretrained on large-scale cross-embodiment data, and Video Prediction Policy (VPP) [33], a lightweight yet effective world-action model that also leverages pretraining. All methods are trained on the same 300 demonstrations per task.

Comparison with baselines. As shown in Table IV, each task is evaluated over 20 trials. DAWN achieves the highest average success rate (72.5%), outperforming GR00T N1.5 (68.3%) and VPP (42.5%). GR00T N1.5 is competitive on individual tasks but lacks explicit future-predictive reasoning; VPP, despite its pretrained world model, does not disentangle dynamics from appearance, limiting effectiveness on visually complex manipulation. These results confirm that distilled structural priors transfer effectively to real-world settings.

Real-world ablation. Table V ablates on the *Stack the Blocks* task, where all variants are trained on the same

TABLE I

EVALUATION RESULTS ON THE LIBERO BENCHMARK. WE REPORT THE TASK SUCCESS RATE (%) FOR EACH SUITE AND THE AVERAGE ACROSS ALL TASKS. EACH TASK IS EVALUATED OVER 50 EPISODES (500 EPISODES PER SUITE).

Method	LIBERO				
	Spatial	Object	Goal	Long	Avg
LAPA [21]	73.8	74.6	58.8	55.4	65.7
UniVLA [22]	96.5	96.8	95.6	92.0	95.2
OpenVLA-OFT [2], [23]	97.6	98.4	97.9	94.5	97.1
π_0 [3]	96.8	98.8	95.8	85.2	94.2
π_0-Fast [24]	96.4	96.8	88.6	60.2	85.5
WorldVLA [25]	87.6	96.2	83.4	60.0	81.8
villa-X [26]	97.5	97.0	91.5	74.5	90.1
GR00T N1 [27]	94.4	97.6	93.0	90.6	93.9
$\pi_{0.5}$ [28]	98.8	98.2	98.0	92.4	96.9
DAWN	96.2	100.0	98.2	94.2	97.2
w/ WM Infer.	96.5	99.7	98.7	94.1	97.3
w/o WM Distill	96.0	99.4	97.4	93.6	96.6

TABLE II

EVALUATION RESULTS ON THE LIBERO-PLUS BENCHMARK. WE REPORT THE AVERAGE SUCCESS RATE (%) ACROSS EACH PERTURBATION DIMENSION, WHERE EACH PERTURBATION INCLUDES THE FOUR TASK SUITES FROM THE ORIGINAL LIBERO BENCHMARK. AVG DENOTES THE AVERAGE SCORE OVER ALL TASKS.

Method	LIBERO-Plus							
	Camera	Robot	Language	Light	Background	Noise	Layout	Avg
UniVLA [22]	1.8	46.2	69.6	69.0	81.0	21.2	31.9	42.9
OpenVLA-OFT [2], [23]	56.4	31.9	79.5	88.7	93.3	75.8	74.2	69.6
π_0 [3]	13.8	6.0	58.8	85.0	81.4	79.0	68.9	53.6
π_0-Fast [24]	65.1	21.6	61.0	73.2	73.2	74.4	68.8	61.6
WorldVLA [25]	0.1	27.9	41.6	43.7	17.1	10.9	38.0	25.0
Fast-WAM [12]	15.9	42.8	66.6	77.8	51.0	37.4	59.0	48.6
DAWN	53.0	68.4	88.2	96.5	97.1	76.3	82.4	78.7
w/ WM Infer.	50.2	70.4	86.3	95.9	97.9	76.1	84.5	78.5
w/o WM Distill	42.9	62.9	84.9	96.3	96.1	74.7	83.0	75.4

TABLE III

EVALUATION RESULTS ON ROBOTWIN 2.0. WE REPORT THE TASK SUCCESS RATE (%) UNDER THE CLEAN SETTING. ALL POLICIES ARE JOINTLY TRAINED ACROSS ALL 10 TASKS. BACKGROUND COLORS INDICATE HORIZON CATEGORY: SHORT (≤ 400), MEDIUM (400–800), AND LONG (≥ 800) STEPS.

Task	DP [29]	ACT [30]	π_0 [3]	RDT [31]	DP3 [32]	DAWN	w/ WM Infer.	w/o WM Distill
Pick Diverse Bottles	6	7	27	2	52	53	55	43
Place Mouse Pad	0	0	7	1	4	33	33	10
Place Object Scale	1	0	10	1	15	66	64	56
Place Object Stand	22	1	36	15	60	84	85	70
Open Laptop	49	56	85	59	82	84	85	75
Place Bread Basket	14	6	17	10	26	80	78	71
Place Object Basket	15	15	16	33	65	82	85	71
Handover Block	10	42	45	45	70	52	53	41
Place Cans Plasticbox	40	16	34	6	48	95	93	65
Stack Blocks Two	7	25	42	21	24	84	84	69
Average	16.4	16.8	31.9	19.3	44.6	71.3	71.5	57.1

300 demonstrations and blocks are deliberately placed at extreme positions to stress-test robustness. The results show a clear progression: the pure action model (w/o WM Distill) achieves only 8/20, adding entangled world model

Task	GR00T N1.5	VPP	DAWN
Insert Drill	10/20	6/20	**11**/20
Hang Towel	13/20	8/20	**15**/20
Chain & Close Box	**16**/20	10/20	**16**/20
Fold Cloth	**14**/20	7/20	13/20
Fold Towel	11/20	5/20	**14**/20
Stack Blocks	**18**/20	15/20	**18**/20
Avg. Success (%)	68.3	42.5	**72.5**

TABLE IV

REAL-WORLD COMPARISON. EACH TASK USES 300 DEMOS, EVALUATED OVER 20 TRIALS. BEST IN **BOLD**.

w/o WM Distill	w/o Disentangle	DAWN
8/20	15/20	**18**/20

TABLE V

REAL-WORLD ABLATION ON *Stack the Blocks* (300 DEMOS, 20 TRIALS AT EXTREME POSITIONS).

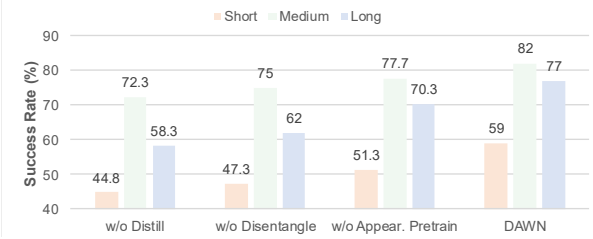


Fig. 5. **Ablation study on RoboTwin 2.0.** Success rates (%) across task horizon categories for four model variants.

Method	Time (ms) ↓
Fast-WAM [12]	442.0
DAWN w/ WM Infer.	148.5
DAWN	**79.6**

TABLE VI

INFERENCE EFFICIENCY. AVG. INFERENCE TIME (MS) PER STEP ON A SINGLE RTX 4090.

distillation (w/o Disentangle) improves to 15/20, and the full DAWN with disentangled distillation reaches 18/20, confirming that both world model distillation and dynamics-appearance disentanglement are essential for robust real-world performance under spatially challenging conditions. Representative successful and failed real-world rollouts are provided in Appendix E-B; see Figures 10 and 11.

D. Analysis

1) Ablation Studies: We ablate components of DAWN on RoboTwin 2.0, as shown in Figure 5, reporting success rates across horizon categories.

Effect of dynamics-appearance disentanglement. Adding the appearance pathway (trained from scratch) yields gains over entangled distillation, especially on long-horizon tasks, validating that a dedicated appearance branch compels the action branch to specialize in dynamics through complementary learning.

Effect of appearance pathway pretraining. Initializing with pretrained InstructPix2Pix weights achieves the strongest performance, as pretrained visual generation priors enable more effective appearance absorption from early training, amplifying disentanglement.

2) Inference Efficiency: Table VI compares inference latency on a single NVIDIA RTX 4090, averaged over 10 runs. DAWN runs at 79.6 ms per step, 1.9× faster than DAWN w/ WM Infer. at 148.5 ms and 5.6× faster than Fast-WAM [12] at 442.0 ms, which still routes observations through the full video DiT backbone as a single-pass encoder. Combined with the competitive task performance in Tables I–III, the results confirm that discarding the world model at inference effectively eliminates generative overhead while preserving the benefits of world model distillation.

IV. CONCLUSION

We present a training-inference asymmetric framework that bridges the gap between WAM effectiveness and real-time deployment efficiency. By distilling action-centric structural priors from a frozen world model through dynamics-appearance disentangled distillation, our approach transfers future-predictive advantages into compact action representations without requiring the world model at inference. Complementary learning with an auxiliary appearance generator ensures the distilled representations focus on action-relevant dynamics rather than visual appearance, enabling video-free deployment that matches WAM-level performance while maintaining standard VLA inference efficiency, and offering a practical path toward deploying world-model-informed policies in real-time closed-loop control.

Limitations and future work. The distilled representations depend on the coverage and quality of the frozen world model; richer contact dynamics, deformable-object interactions, and highly novel scenes may benefit from stronger teachers or additional sensory supervision. Future work could extend this framework with uncertainty-aware distillation, tactile feedback, and continual closed-loop refinement, enabling the action model to adapt through real physical interaction.

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APPENDIX A
RELATED WORK

A. Vision-Language-Action Models

The remarkable generalization achieved by large-scale vision-language pretraining has motivated researchers to repurpose such multimodal backbones for robotic control, giving rise to the Vision-Language-Action (VLA) paradigm—end-to-end models that map raw visual observations and language instructions directly to motor commands. RT-2 [1] pioneered this direction by demonstrating that a pretrained vision-language model can be fine-tuned to output tokenized robot actions, thereby transferring web-scale knowledge to manipulation tasks. This insight was rapidly amplified by open-source generalist policies such as OpenVLA [2] and Octo [34], which leveraged large cross-embodiment datasets [35]–[37], and by the π_0 family [3], [28], [38] that progressively scaled flow-matching VLAs toward open-world generalization and continual learning from online experience. In parallel, industrial foundation models [27], [39]–[43] and multimodal agent architectures [44]–[46] have driven the paradigm toward real-time deployment and unified perception–action capabilities, supported by a growing ecosystem of open designs [22], [31], [47]–[54] evaluated on standardized benchmarks [18], [19], [55]–[59].

A central design question in VLAs is how to represent continuous robot actions within a discrete language-model backbone. One strand of research develops compact discrete tokenizers—via DCT-based compression [24], [60], [61], vector quantization [62], or universal latent action distillation [63], [64]—that enable autoregressive generation of multi-step action chunks. A complementary strand replaces discrete decoding entirely with continuous diffusion or flow-matching action heads [65]–[71], allowing multimodal action distributions to be captured without discretization artifacts. Hybrid architectures that combine autoregressive reasoning with diffusion-based action generation [72], [73] further suggest that the two paradigms offer complementary strengths, and the optimal design trade-off remains an open question.

Reactive, single-step policies frequently struggle with compositional or long-horizon manipulation, motivating the integration of explicit or latent reasoning into VLAs. Explicit chain-of-thought methods produce visual or linguistic rationales prior to acting [74]–[80], whereas latent reasoning approaches encode deliberation into hidden representations for faster inference [81]–[88]. Orthogonally, monocular VLAs often misjudge depth and contact geometry, prompting the integration of point clouds, stereo cues, and explicit 3D representations [32], [89]–[104]. Because both reasoning and 3D perception incur additional computational cost, a parallel thread investigates making VLAs lightweight enough for real-time control through layer skipping, token pruning, early-exit decoding, and knowledge distillation [105]–[115].

Beyond core architecture design, extending VLA capabilities along multiple axes has attracted broad attention. On the input side, force/torque and tactile sensors supply contact-rich feedback that vision alone cannot capture [116]–[118], while recurrent and reconstructive memory mechanisms enable history-aware perception over extended manipulation sequences [119]–[123]. On the data side, learning from large-scale egocentric and third-person human videos bridges the embodiment gap between web-sourced data and robot demonstrations [64], [124]–[127]. On the embodiment side, cross-platform transfer [128]–[130] and morphology-specific architectures for whole-body locomotion [131], mobile manipulation [132], bimanual coordination [133], and dexterous hands [134], [135] extend VLAs well beyond single-arm table-top settings. Task-level generalization has been pursued through visual prompting interfaces [136]–[138], handling of dynamic objects and task switching [139]–[141], richer instruction following [142]–[144], and hierarchical planning for long-horizon goals [145]–[151].

Although imitation learning dominates VLA training, its performance ceiling is bounded by the quality and coverage of demonstrations. Reinforcement learning offers a principled route to surpass this ceiling through reward-verified fine-tuning in simulation [152]–[157], test-time verification and best-of-N sampling [158]–[162], and systematic empirical studies that characterize when and how RL enhances VLA generalization [23], [163]–[168]. Complementary adaptation strategies that steer pretrained VLAs toward novel scenarios without full retraining have also shown promise [169]–[172].

Most relevant to our work, a growing body of VLA methods enriches the action policy with a predictive world model. These approaches learn next-state or next-frame predictions as auxiliary objectives [173]–[181], close the loop between world-model imagination and policy improvement [156], [182]–[186], or decouple intent prediction from action execution [187], [188]. Recent diagnostics also test whether action-conditioned world models faithfully follow the supplied actions [189]. While these methods demonstrate that world knowledge benefits action prediction, they typically *jointly* train the predictive and control components, coupling their representational capacity and optimization dynamics. In contrast, our framework first trains a world model in isolation to master state-transition dynamics, then distills the learned representations into a lightweight action policy—allowing each module to specialize without mutual interference.

B. World Action Models

The finding that video generation models trained on internet-scale data implicitly capture physical dynamics has spurred a family of methods that employ video prediction as a planning substrate. A recent survey organizes this expanding world-action-model design space [190]. Pioneering work demonstrated that text-conditioned video synthesis can serve as a universal policy planner [9], [10], and subsequent methods extracted control signals from actionless video via dense correspondences [191] or leveraged generated human-hand videos to guide robot execution [192], [193]. Scaling this idea, large video foundation models have been pretrained on web-scale data and then adapted for manipulation [7], [8], [11], [194]–[197], with extensions to long-horizon video generation [180], [198], task-relevant attention specialization [199], and multi-modal video–action co-generation [200], [201].

A key limitation of cascaded video-then-action pipelines is that errors in the predicted video compound before reaching the action decoder. Joint world-action models (WAMs) address this by simultaneously denoising or generating video and actions within a shared architecture, enabling bidirectional information flow between predictive and control representations [4]–[6], [202]–[207]. This paradigm has been extended to unified autoregressive formulations [25], [208]–[210], action-centered architectures [211], [212], implicit value-based planning [149], [213], [214], future representation alignment [215]–[217], and multi-modal world modeling that incorporates tactile feedback [218]. Remarkably, recent results show that sufficiently powerful WAMs can serve as zero-shot robot policies without any task-specific training [219], [220], producing predictive visual representations that are themselves strong generalist controllers [33].

Operating in pixel space, however, is computationally expensive and entangles visual appearance with the underlying dynamics. Latent action pretraining [21], [103], [221] addresses this by learning a compact action space from unlabeled video, enabling large-scale VLA training without paired action labels. This principle has been extended to adaptable latent world models [26], [130], [222], [223], billion-parameter latent dynamics modules [224], and skill-compositional post-training [225]–[229]. Concurrently, 4D spatiotemporal world models [230]–[232] capture geometric evolution over time. A growing line of work also addresses computational and deployment challenges: some investigate whether test-time imagination is always necessary or can be bypassed via distilled shortcuts [12], [13], [112], [162], [233]–[236], others explore online adaptation to changing conditions [172], [237], [238], and recent efforts enrich WAM representations through object-level structure [113], [239], intent-aware spatial value maps [240], [241], parameter-efficient cross-model knowledge transfer [242], and system-level integration of multiple predictive components [243], [244]. ActiveWAM further couples observation selection with world-action modeling [241].

Recent surveys summarize the broader language-conditioned manipulation and world-model landscape [245], [246]. Despite the rapid progress in both VLA and WAM communities, current approaches share a common architectural commitment: world modeling and action generation are tightly coupled within a single training loop, forcing both modules to share representations and optimization schedules. Our work challenges this premise. By first training a world model to learn rich, transferable state-transition dynamics and then distilling this knowledge into a dedicated action policy, we allow each module to specialize in its own objective, yielding a decoupled framework that combines the predictive power of world models with the deployment efficiency of lightweight VLA policies.

APPENDIX B
WORLD MODEL DETAILS

A. Architecture

The backbone is built on LTX-Video [14], a 2B-parameter Diffusion Transformer (DiT) pretrained for video generation. Its video VAE provides $16\times$ spatial downsampling and $8\times$ temporal downsampling, substantially reducing the token count for efficient training and inference. Given a language instruction and a variable-view history frame, the backbone predicts a future clip at 256×256 resolution over a horizon of $K=24$ frames using a flow-matching objective in the VAE latent space.

B. Training Data and Configuration

The video diffusion backbone is adapted on a mixture of robot manipulation videos and egocentric human activity data:

- *Robot videos*: Open X-Embodiment [35], AgiBot-World [37], Galaxea [247], RoboMIND [248], and RoboCOIN [249], collectively spanning single-arm and bimanual platforms with diverse end-effectors, camera viewpoints, and task categories.
- *Human videos*: Ego4D [250] and Humanoid Everyday [251], providing rich egocentric interaction patterns including object manipulation, tool use, and dexterous hand activities.

The combined video corpus covers a broad spectrum of embodiments, interaction dynamics, and visual environments, enabling the backbone to internalize generalizable spatiotemporal priors. During DAWN training, the video backbone is kept entirely frozen. Only the distillation module and the downstream action head are optimized, ensuring that the world model's pretrained spatiotemporal priors are transferred without degradation.

For downstream evaluation, the action model is fine-tuned on benchmark-specific demonstration data: LIBERO training demonstrations for both LIBERO and LIBERO-Plus evaluation, and the RoboTwin 2.0 clean-setting demonstrations for RoboTwin 2.0 evaluation.

APPENDIX C
DAWN ARCHITECTURE AND TRAINING DETAILS

A. Model Architecture

Table VII summarizes the architecture of each component in DAWN. The configuration shown corresponds to the RoboTwin 2.0 training setup.

B. Training Hyperparameters

Table VIII lists the training hyperparameters used for DAWN on the RoboTwin 2.0 benchmark.

APPENDIX D
REAL-WORLD EXPERIMENTAL SETUP

Figure 6 shows the two real-world robot embodiments used in our physical experiments, including the head-mounted and wrist-mounted RGB camera views. Figure 7 summarizes the initial and goal configurations for the six manipulation tasks evaluated in the real world.

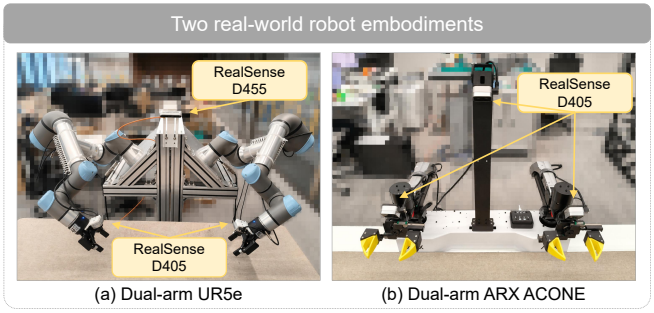


Fig. 6. **Real-world robot embodiments.** Left: the dual-arm UR5e setup with a head-mounted RealSense D455 and wrist-mounted RealSense D405 cameras. Right: the dual-arm ARX ACONE setup with the same multi-view camera configuration.

APPENDIX E
QUALITATIVE VISUALIZATION

A. Simulation Rollout Visualization

Figure 8 presents representative successful rollouts of DAWN across all three simulation benchmarks: LIBERO, LIBERO-Plus, and RoboTwin 2.0. These qualitative examples illustrate the diversity of manipulation skills acquired by our policy, including precise spatial reasoning, robust generalization under domain perturbations, and coordinated bimanual control over varying task horizons. Figure 9 shows a representative failure case on RoboTwin 2.0, where the policy fails to coordinate simultaneous dual-arm bottle grasping, indicating insufficient coverage of symmetric bimanual synchronization.

B. Real-World Rollout Visualization

Figure 10 presents representative successful real-world rollouts across the six physical tasks. Figure 11 shows a representative failure case on towel folding, where the policy makes partial progress but does not complete the final fold.

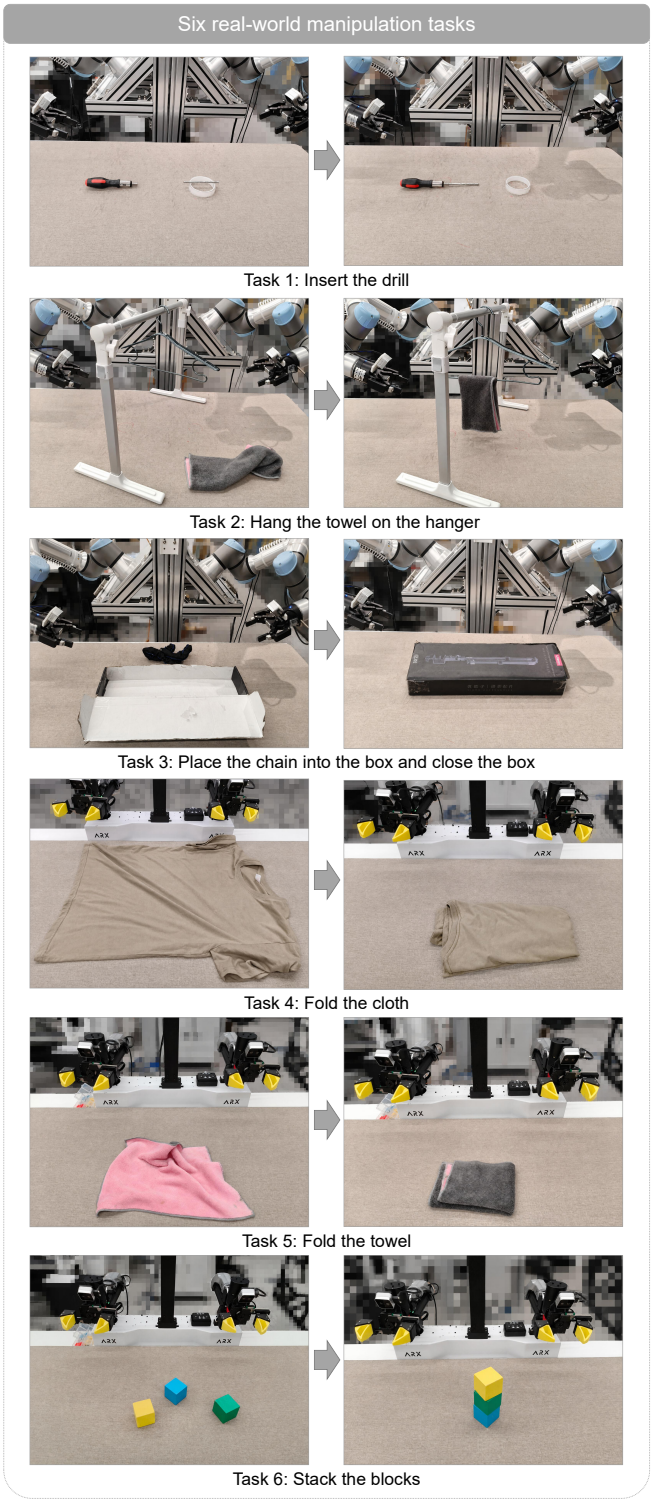


Fig. 7. **Real-world manipulation tasks.** Initial and goal configurations for six tasks: Insert the Drill, Hang the Towel on the Hanger, Place the Chain into the Box and Close the Box, Fold the Cloth, Fold the Towel, and Stack the Blocks.

TABLE VII

MODEL ARCHITECTURE OVERVIEW.

Component	Specification	Parameters
VLM backbone	Qwen3-VL-4B-Instruct	4B
Hidden dimension	2048	—
Observation resolution	224×224	—
Action head	MLP	—
Hidden size	1024	—
Action chunk length	50	—
Action dimension	14	—
World model backbone	LTX-Video DiT (frozen)	2B
Num layers	28	—
Input resolution	256×256	—
Future horizon	24 frames	—
Appearance pathway	InstructPix2Pix + LoRA	1B
Feature decoder	Conv2d + GroupNorm + Conv2d	—

TABLE VIII

TRAINING HYPERPARAMETERS.

Hyperparameter	Value
Optimizer	AdamW
Learning rate (VLM backbone)	1×10^{-5}
Learning rate (action head)	1×10^{-4}
Learning rate (appearance pathway)	5×10^{-6}
LR scheduler	Cosine (min LR 5×10^{-7})
Warmup steps	5,000
Batch size (per device)	16
Gradient accumulation steps	6
Distributed strategy	DeepSpeed ZeRO-2
GPUs	2× NVIDIA H200

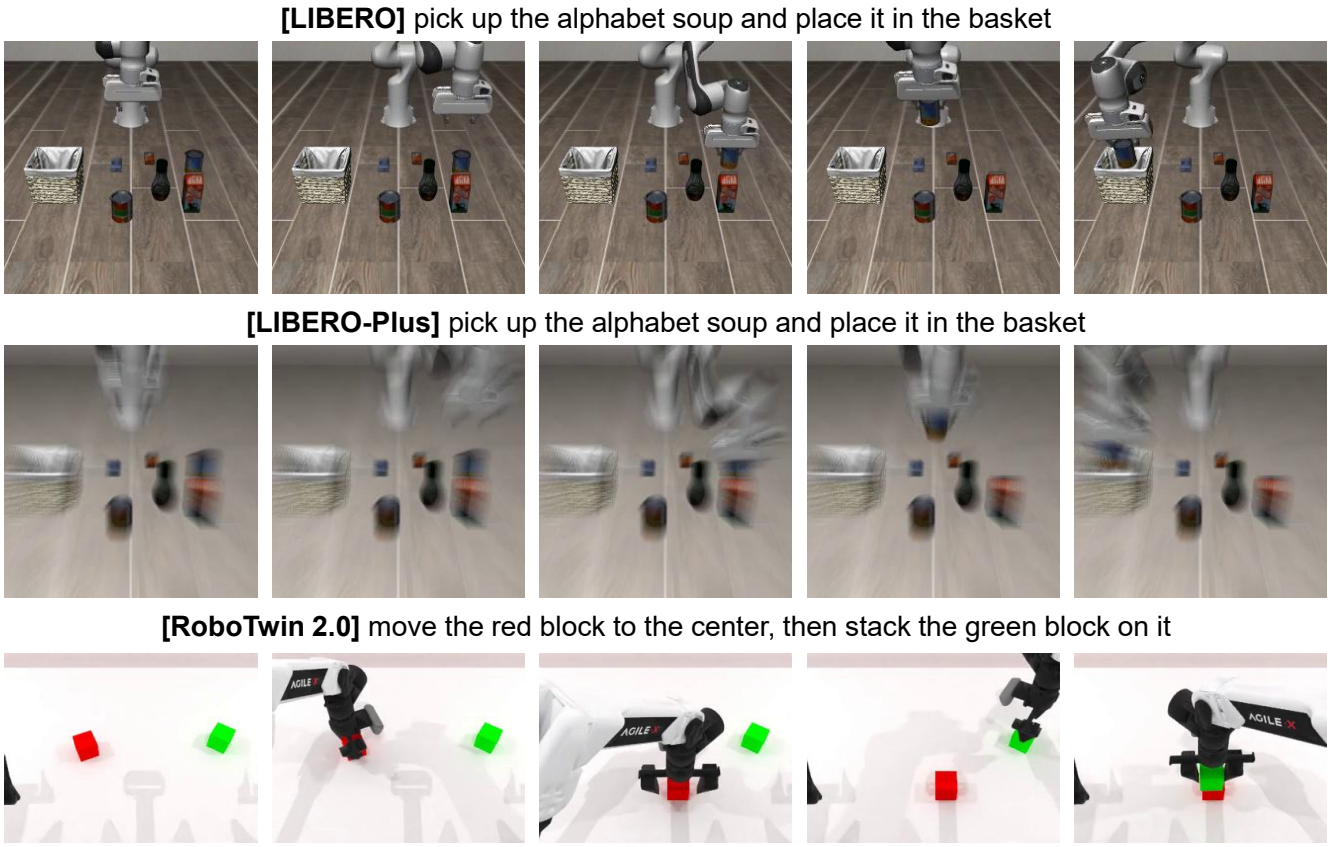


Fig. 8. **Qualitative visualization of successful rollouts on three benchmarks.** Each row shows keyframe sequences from representative tasks in LIBERO, LIBERO-Plus, and RoboTwin 2.0, demonstrating the diverse manipulation capabilities of DAWN.



Fig. 9. **Representative simulation failure case on RoboTwin 2.0.** A bimanual task in which the policy fails to coordinate simultaneous dual-arm grasping.



Fig. 10. **Representative successful real-world rollouts.** Keyframe sequences from six physical manipulation tasks demonstrate that DAWN transfers to insertion, hanging, box closing, deformable-object folding, and block stacking across the two real-world embodiments.



Fig. 11. **Representative real-world failure case.** A towel-folding rollout in which the policy grasps and moves the towel but does not complete the final fold, illustrating a remaining failure mode under deformable-object manipulation.